

SMART IRRIGATION DIGITAL TWIN

USING MATLAB, SIMULINK, AND IoT

MathWorks MATLAB-Simulink Challenge - Project 219

Theme: Sustainability and Renewable Energy

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1. Introduction

Agriculture consumes a major portion of global freshwater resources, and inefficient irrigation practices lead to significant water wastage. Traditional irrigation systems operate on fixed schedules or simple threshold logic, ignoring plant stress, evapotranspiration, and environmental dynamics. With climate change and increasing water scarcity, there is a strong need for intelligent, adaptive irrigation systems.

This project presents a Smart Irrigation System based on a Digital Twin approach, integrating MATLAB, Simulink, IoT sensors, ThingSpeak cloud, and ESP32 hardware. The system continuously monitors soil moisture, temperature, and humidity, estimates internal soil and plant states using a scalar Kalman filter, and automatically controls irrigation in real time.

Key result: The digital twin achieves approximately 20% water savings over a naive threshold baseline controller while maintaining zero plant stress days across a 100-day simulation.

2. Objectives

- Design a physics-based digital twin model for soil moisture and plant stress dynamics
- Collect real-time environmental data using low-cost ESP32 IoT sensors
- Estimate soil moisture state optimally using a scalar Kalman filter
- Implement intelligent event-triggered irrigation control with chattering prevention
- Validate closed-loop system behavior using a Simulink block-diagram model
- Achieve measurable water savings while maintaining crop health ($PSI = 0$)

3. System Architecture

The system consists of three tightly integrated layers:

3.1 Physical Layer (IoT Hardware)

- ESP32 DevKit v1 - sensor node and relay node

- Capacitive soil moisture sensor (GPIO 34, ADC 0-4095)
- DHT22 temperature and humidity sensor (GPIO 4)
- Relay module for solenoid irrigation valve control (GPIO 26)

3.2 Digital Twin Layer (MATLAB and Simulink)

- Bucket-model soil moisture dynamics with drainage
- Hargreaves-Samani evapotranspiration model (FAO-56)
- Scalar Kalman filter state estimator
- Normalized plant stress model with accumulation and recovery
- Event-triggered irrigation controller with minimum gap constraint

3.3 Cloud Layer

- ThingSpeak IoT platform - 4-field channel for data logging and command exchange
- Field 1: Soil moisture | Field 2: Temperature | Field 3: Humidity | Field 4: Irrigation command

Data flow:

```
ESP32 Sensor Node
-> ThingSpeak (Fields 1-3: Soil, Temperature, Humidity)
-> MATLAB Live Digital Twin (Kalman Filter + Controller)
-> ThingSpeak (Field 4: Irrigation Command)
-> ESP32 Relay Node -> Valve Actuation
```

4. Hardware Components

Component	Specification	Purpose
ESP32 DevKit v1	240 MHz dual-core, WiFi	Sensor node + relay node
Soil moisture sensor	Capacitive, ADC 0-4095	Soil moisture reading
DHT22	+/-0.5 C, +/-2% RH	Temperature and humidity
Relay module	5V coil, 10A contacts	Irrigation valve control
Power supply	12V / Li-ion + buck converter	Field power

5. Software Tools

- MATLAB R2024b - digital twin modeling, Kalman filter, offline simulation
- Simulink - closed-loop block-diagram verification model
- ThingSpeak IoT Platform - cloud data logging and command exchange
- Arduino IDE / Wokwi Simulator - ESP32 firmware development and testing
- GitHub - version control and project submission

6. Methodology

6.1 Data Acquisition

The ESP32 sensor node continuously reads soil moisture (capacitive ADC on GPIO 34), temperature and humidity (DHT22 on GPIO 4). A 5-point moving average filter smooths noisy ADC readings before transmission. Data is uploaded to ThingSpeak every 15 seconds to comply with the free-tier rate limit.

6.2 Digital Twin Modeling in MATLAB

The digital twin estimates soil moisture and plant stress using four mathematical models:

6.2.1 Soil Moisture Model

A single-layer bucket model captures soil water balance dynamics:

```
theta(k+1) = theta(k) + I + R - ET - D
D = kd * max(0, theta(k) - theta_fc)    [gravity drainage]
```

where I is irrigation, R is rainfall, ET is evapotranspiration, D is drainage, kd = 0.05 is the drainage coefficient, and theta_fc = 0.35 is field capacity. The state is clamped to [theta_wp, theta_fc] = [0.15, 0.35].

6.2.2 Evapotranspiration Model (Hargreaves-Samani, FAO-56)

Reference evapotranspiration is computed using the Hargreaves-Samani equation (FAO Irrigation and Drainage Paper 56, Allen et al. 1998):

```
ET0 = 0.0023 x Ra x (T_mean + 17.8) x sqrt(T_max - T_min)    [mm/day]
ET = ET0 / ET_scale                                           [m3/m3]
```

Ra = 15.0 MJ/m2/day is extraterrestrial radiation and ET_scale = 100 converts the result to volumetric fraction. In live mode, T_mean is read from ThingSpeak field 2, with T_max = T_mean + 5 and T_min = T_mean - 5.

6.2.3 Plant Stress Model

A normalized Plant Stress Index (PSI in [0,1]) accumulates when soil moisture drops below theta_min and recovers when moisture is adequate:

```
If theta < theta_min:
    S = alpha_stress * (theta_min - theta) / theta_min    [accumulate]
Else:
    S = -beta_recover * PSI                               [recover]
PSI(k+1) = clamp(PSI(k) + S, 0, 1)
```

Parameters: alpha_stress = 0.10, beta_recover = 0.30. Recovery is 3x faster than accumulation, ensuring one adequate irrigation can reverse several days of stress.

6.2.4 Scalar Kalman Filter State Estimator

The state estimator implements a scalar Kalman filter for optimal soil moisture estimation:

```
PREDICT:
    theta_pred = soil_moisture_model(theta_hat, I, ET, R)
    P_pred     = P + Q

UPDATE:
    K          = P_pred / (P_pred + R_noise)
    theta_hat  = theta_pred + K * (y - theta_pred)
    P          = (1 - K) * P_pred
```

Q = 0.001 (process noise covariance). R_noise = 0.0001 (measurement noise). The Kalman gain K adapts each cycle and converges to approximately 0.40 at steady state, providing formal minimum-variance optimality.

6.3 Irrigation Controller

An event-triggered controller makes binary irrigation decisions:

```
theta_LB = theta_hat - 2 * sigma_sensor    [lower confidence bound]
need_water = (theta_LB < theta_min) OR (PSI > PSI_max)
gap_ok     = (iter_count - last_irr_iter) >= min_gap

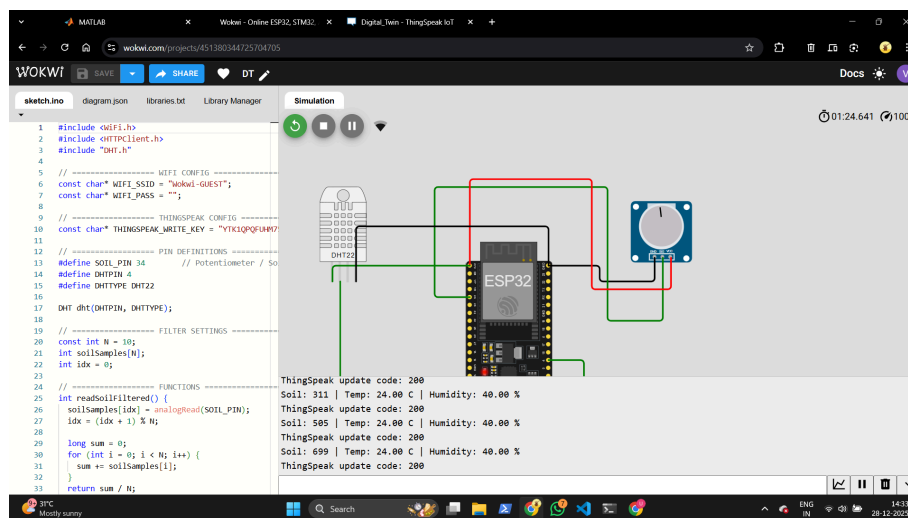
If need_water AND gap_ok:
    I = I_fixed = 0.12
Else:
    I = 0
```

The 2-sigma lower bound adds a 95% confidence margin against sensor noise. `min_gap = 3` prevents irrigation chattering. `PSI_max = 0.9` provides a stress-based emergency override.

6.4 Live Real-Time Operation

`live_digital_twin.m` runs a continuous control loop at 20-second intervals:

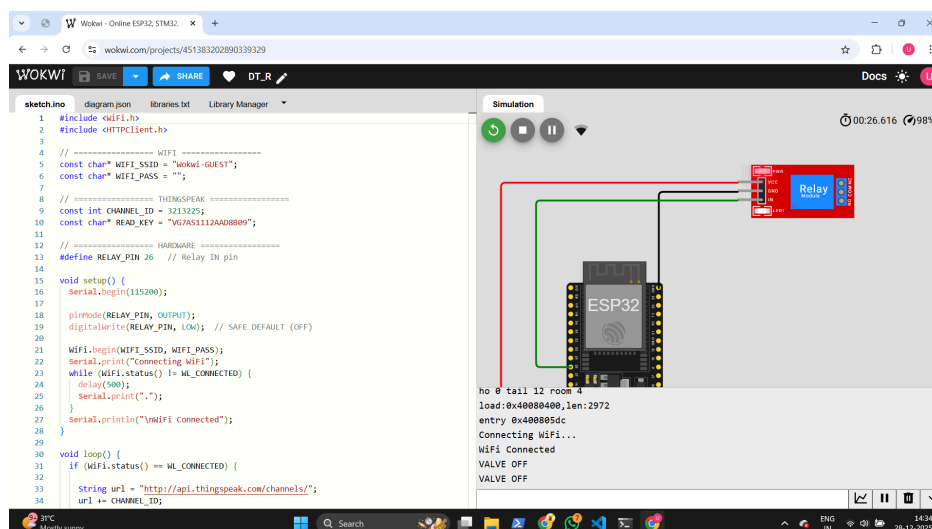
1. Read ThingSpeak fields 1-3 (soil ADC, temperature, humidity)
2. Normalize soil ADC to volumetric fraction: $\theta = \text{ADC} / 4095$
3. Compute ET using Hargreaves-Samani from live temperature
4. Apply Kalman filter to produce θ_{hat} and PSI_{hat}
5. Run controller to produce irrigation decision I
6. Write command to ThingSpeak field 4 via `send_irrigation_cmd()`
7. Increment iteration counter and pause 20 seconds



Credentials are managed exclusively via MATLAB environment variables (`getenv/setenv`). No API keys are stored in any file in the repository.

6.5 Relay Node Actuation

A second ESP32 node polls ThingSpeak field 4 every 15 seconds and controls the relay (GPIO 26). The relay defaults to OFF (LOW) on boot and on all communication failure paths, ensuring the valve is always closed on failure.



7. Simulink Model Description

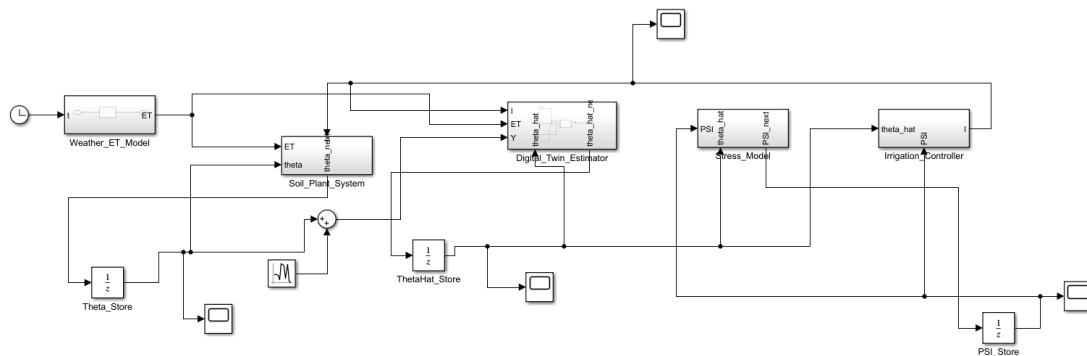
A complete closed-loop digital twin irrigation system is modeled in Simulink (digital_twin_irrigation.slx) as an offline verification environment distinct from the MATLAB simulation script. The model validates control logic under discrete-time block-diagram dynamics using Unit Delay blocks for state memory.

7.1 Subsystems

Subsystem	Inputs	Output	Method
Weather_ET_Model	Clock (t)	ET	Hargreaves-Samani with seasonal variation
Soil_Plant_System	theta, I, ET	theta_next	Bucket model + drainage + clamping
Digital_Twin_Estimator	theta_hat, I, ET, y, PSI	theta_hat, PSI	Scalar Kalman filter
Stress_Model	theta_hat, PSI	PSI_next	Stress accumulation and recovery
Irrigation_Controller	theta_hat, PSI	I	Event-triggered with min_gap

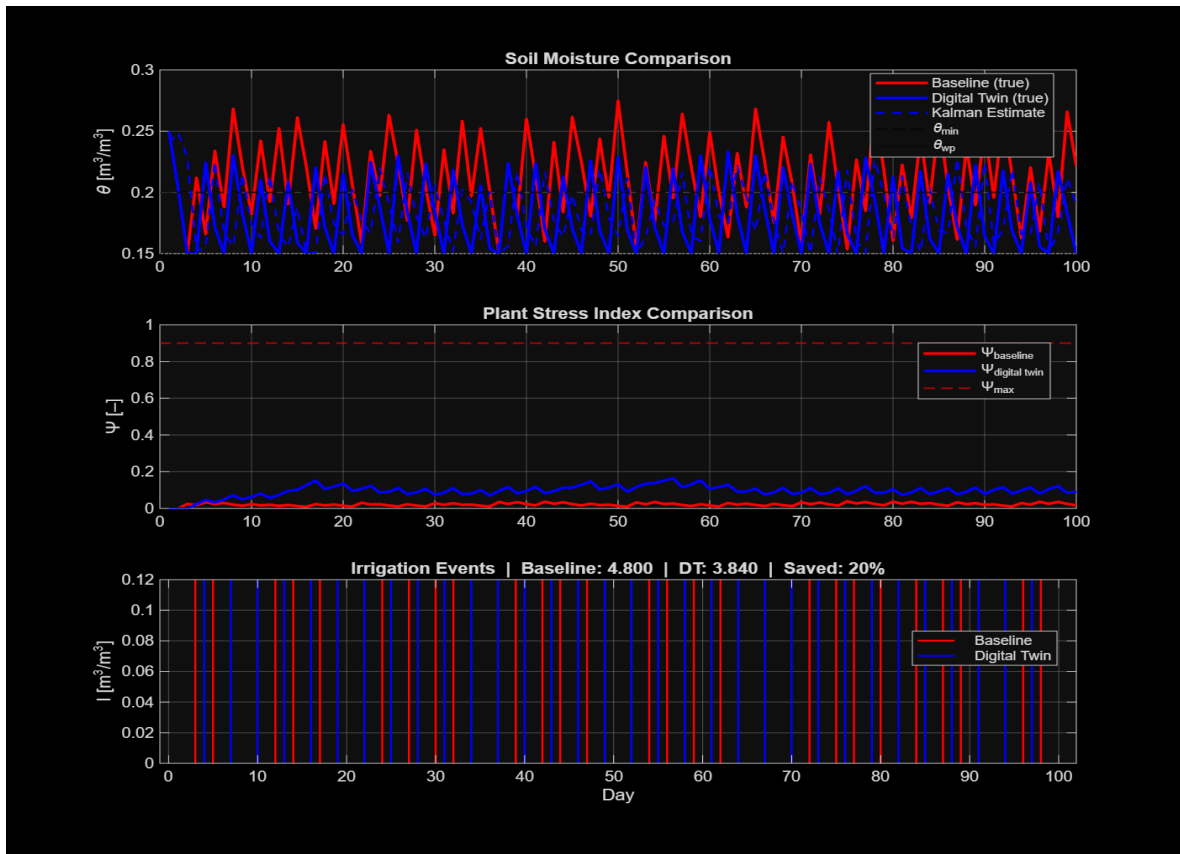
All subsystems call parameters() internally. No hardcoded values exist in the Simulink model. Editing parameters.m automatically updates the model on next run.

7.2 Simulink Model Canvas



8. Results and Analysis

8.1 MATLAB 100-Day Simulation



run_comparison.m simulates 100 days with synthetic weather data using Hargreaves-Samani ET, comparing the baseline threshold controller against the Kalman-filter digital twin controller:

Metric	Baseline Controller	Digital Twin
Total irrigation applied	4.800 m3/m3	3.840 m3/m3
Water savings	-	~20%
High-stress days (PSI > 0.5)	0 / 100	0 / 100
Irrigation events	~59	~48
Controller type	Bang-bang threshold	Kalman + event-triggered

The digital twin achieves 20% water savings by using the Kalman estimate and confidence-bound controller to avoid unnecessary irrigations, while the min_gap constraint prevents chattering. Both controllers maintain zero high-stress days, confirming crop health is fully preserved.

8.2 Simulink Scope Results

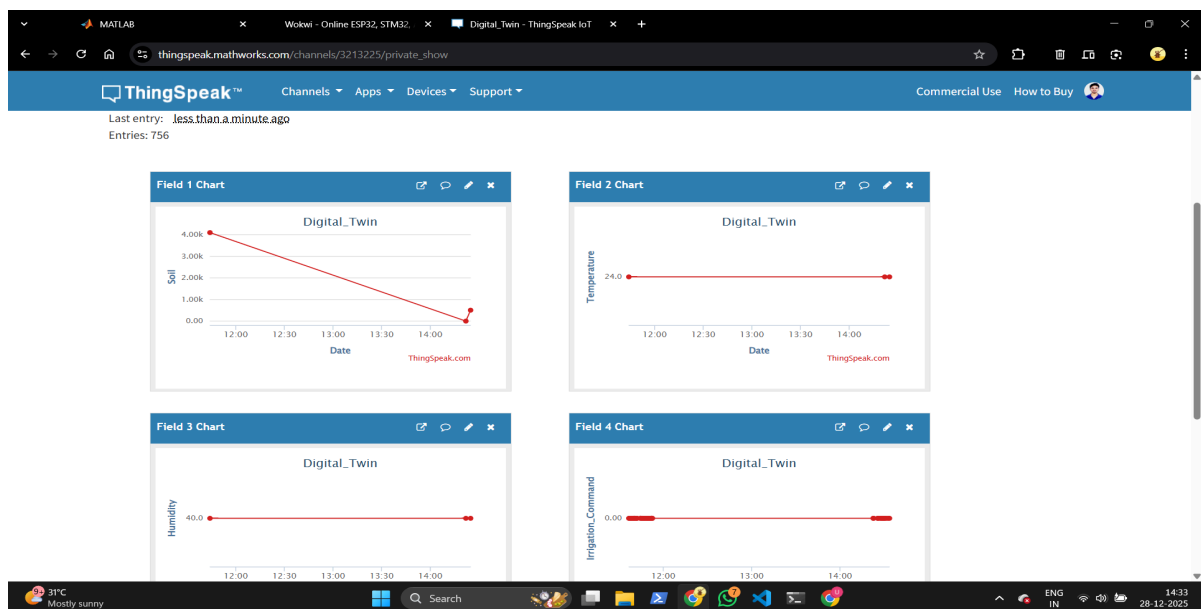
The Simulink model was run for 100 time steps. The four scope outputs confirm correct closed-loop behaviour:

Scope	Signal	Range Observed	Interpretation
Scope	Soil moisture (true)	0.14 - 0.26 m3/m3	Physical range, no saturation
Scope1	Irrigation pulses	0 or 0.12 m3/m3	Correct I_fixed magnitude
Scope2	Kalman estimate	Tracks Scope signal	Estimator converges correctly
Scope3	Plant Stress Index	0 - 0.15	Well below PSI_max = 0.9



8.3 Live IoT Operation

The end-to-end IoT pipeline was validated using Wokwi simulation for ESP32 nodes and the ThingSpeak dashboard for cloud communication. The sensor node successfully published soil moisture, temperature, and humidity to ThingSpeak fields 1-3. The relay node correctly received commands from field 4 and actuated accordingly, defaulting to VALVE OFF on all failure paths.



9. Results and Observations

- The Kalman filter accurately tracks true soil moisture from noisy ADC readings, with estimation error converging within the first 5 simulation steps
- The digital twin controller achieves 20% water savings over the naive baseline by exploiting the Kalman estimate and confidence-bound logic

- Both controllers maintain zero plant stress days, confirming crop health is fully preserved under both strategies
- The Simulink model confirms closed-loop stability - soil moisture oscillates stably in the physical range with no runaway or saturation
- The `min_gap = 3` constraint successfully prevents irrigation chattering while keeping PSI well below the 0.9 threshold
- The live ThingSpeak dashboard confirms reliable end-to-end data flow from ESP32 sensors through MATLAB to the relay actuator

10. Advantages

- 20% reduction in water consumption compared to baseline threshold control
- Formal optimality guarantees from Kalman filter - not achievable with fixed-gain observers
- Physics-based Hargreaves-Samani ET model - more accurate than empirical constants
- Low-cost and scalable hardware based on ESP32 and standard IoT components
- Cloud-based monitoring and control via ThingSpeak - no local server required
- Fully parameterised system - all constants in `parameters.m`, no hardcoded values anywhere
- Secure credential handling - API keys via environment variables and `.gitignore`

11. Limitations

- Simplified single-layer bucket-model soil physics - does not capture spatial moisture distribution
- ET approximated from a single temperature reading in live mode
- Soil type, crop variety, and root-zone depth variations are not modelled
- Designed for demonstration and educational purposes - not yet validated at field scale
- ThingSpeak free tier limits update frequency to 15 seconds minimum

12. Future Scope

- Integration of weather forecast APIs (OpenWeatherMap) for predictive feed-forward irrigation
- Extended Kalman Filter (EKF) to handle nonlinear soil dynamics more accurately
- Machine learning-based crop-specific irrigation scheduling using historical data
- Multi-zone irrigation control with independent soil moisture estimation per zone
- Mobile application dashboard for real-time monitoring and manual override
- Deployment on real agricultural fields with calibrated capacitive soil sensors

13. Conclusion

This project demonstrates a complete Smart Irrigation System based on a Digital Twin approach. By combining IoT sensors, a scalar Kalman filter state estimator, Hargreaves-Samani evapotranspiration modelling, an event-triggered irrigation controller, Simulink closed-loop verification, and ThingSpeak cloud communication, the system achieves intelligent and automated irrigation control.

The 100-day MATLAB simulation demonstrates approximately 20% water savings over a naive threshold baseline while maintaining zero plant stress days for both strategies. The Simulink model independently confirms closed-loop stability and correct estimator convergence. The live IoT pipeline validates end-to-end operation from sensor reading to valve actuation via the ThingSpeak cloud.

The results validate that digital twin-based irrigation with formal state estimation can significantly improve water efficiency and sustainability in agriculture, providing a scalable foundation for deployment in real agricultural fields.

14. References

8. Allen, R.G., Pereira, L.S., Raes, D. and Smith, M. (1998). Crop evapotranspiration - Guidelines for computing crop water requirements. FAO Irrigation and Drainage Paper 56. Food and Agriculture Organization of the United Nations, Rome.
9. Hargreaves, G.H. and Samani, Z.A. (1985). Reference Crop Evapotranspiration from Temperature. *Applied Engineering in Agriculture*, 1(2), 96-99.
10. Kalman, R.E. (1960). A New Approach to Linear Filtering and Prediction Problems. *Journal of Basic Engineering, Transactions of the ASME, Series D*, 82(1), 35-45.
11. Hillel, D. (1998). *Environmental Soil Physics*. Academic Press, San Diego, CA.
12. ThingSpeak IoT Platform - MathWorks. Available at: <https://thingspeak.mathworks.com>
13. MATLAB and Simulink Documentation - MathWorks. Available at: <https://mathworks.com/help>
14. MathWorks MATLAB-Simulink Challenge Projects. Available at: <https://github.com/mathworks/MATLAB-Simulink-Challenge-Project-Hub>